Classification of Self-Actions and the Unique Involutive Survivor

Abstract

We classify self-actions $a : \alpha \to \alpha$ by local orbit shape. Rest, asymmetric step, and erasure each meet a formal obstruction in the ambient of pointed symmetric steps; the surviving shape is the fixed-point-free two-cycle. That survivor is characterised by a universal property: $(\mathsf{Bool}, \neg)$ is initial among pointed unoriented acts, and every fundamental act reads as negation. From fundamentality we obtain a Lawvere-style seal with quantitative grain (names lag predicates at every level, and the lag recedes under completion), an ω-ladder of namers whose colimit is duration, and, after transferring losslessness to forward microsteps by two-bounce factorisation, registration, capacity, and a single $\mathbb{Z}/2\mathbb{Z}$-underdetermination with three faces. Canonicity of the two-bounce pair fails; channel *swap* nevertheless covers step reversal, and content gauges are separated from packaging gauges so that factorisation non-uniqueness does not inflate the dictionary $\mathbb{Z}/2\mathbb{Z}$. On this base we assemble a conditional monist package, *self-articulation*: a registered denial of one drawn distinction draws the registering carrier (retorsion); a drawn distinction forces the live involutive reading; witnesses arise as ladder inhabitants; every presentation of the canon to a second term retracts the canon's own articulation; and completeness exists only as duration. The sole worked dictionary exhibit is a certificate for the meta-problem of consciousness, deriving the report structure of the explanatory gap while refusing all qualia claims. The development is machine-checked in Lean 4 (no Mathlib, no `sorry`).

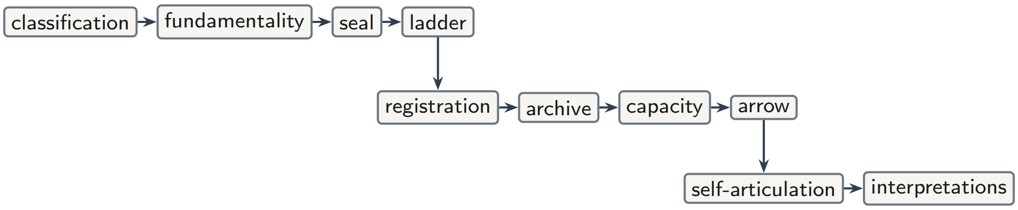
Keywords. classification; involution; universal property; Lawvere diagonal; self-articulation; retorsion; meta-problem; underdetermination; gauge; machine-checked mathematics.

1 Introduction

The central contribution of this paper is an *eliminative classification of self-actions*. Given $a : \alpha \to \alpha$, the local possibilities at a state are rest, two-cycle, asymmetric step, and—as a pairwise defect—erasure. In a precisely specified ambient, every shape except one encounters an obstruction. The survivor is the live two-cycle. Negation on Bool [4] is not assumed at the outset; it is the unique object (up to label swap) satisfying the universal property, in the standard categorical sense [19], of that survivor.

A second contribution assembles, on that base, a conditional monist package. Given one drawn distinction, articulation as negation is forced, witnesses arise only as inhabitants of the resulting ladder, no presentation of the canon to a second term carries content beyond the canon's own articulation, and completeness exists only as the colimit of ascent. The package is a single machine-checked conjunction (Theorem 11.7); its identification with worldhood is confined to a marked interpretation, and the sole worked dictionary exhibit is a certificate for the meta-problem of consciousness.

The formal development then follows a fixed dependency:



Each arrow is a theorem or a defined interface; philosophical commentary is postponed until the supporting machinery is in place.

Method and layers. The text maintains three layers.

(1) **Formal mathematics** — definitions, theorems, proofs.

(2) **Philosophical interpretation** — what the mathematics suggests, marked as such.

(3) **Dictionary** — optional physical or contemplative identifications, admitted only by certificate.

Claims are tagged by status: proved, corollary, interpretation, open, refused, or philosophical residue. Residue is reduced by reclassification: items become theorems, audited residues at the base, or certificate data (Section 13). The companion Lean 4 corpus [9] (no Mathlib [20], no `sorry`) supplies declaration names; this paper is self-contained at the level of statements.

Roadmap. Section 2 gives the architecture. Sections 3–4 carry out the classification and the universal characterisation of the survivor. Sections 5–6 obtain seal and ladder. Sections 7–9 transfer losslessness, record registration and capacity, and identify the arrow with address. Section 10 states the single $\mathbb{Z}/2\mathbb{Z}$ and the gauge demarcation. Section 11 assembles the self-articulation package and sorts its admissible readings. Section 12 confines the dictionary to one worked certificate. Section 13 collects open, refused, and philosophical items. Section 14 situates the development in prior work. Section 15 concludes.

2 Architecture

The companion corpus is stratified into three layers with downward-only imports (Figure 1).

Within the foundation, the dependency is classification $\to$ fundamentality $\to$ seal $\to$ ladder. The dynamics layer imports losslessness along an interface (registration via two-bounce), realises archive and capacity on the ladder carrier, and exports address as the arrow reading. The dictionary layer consumes those exports; it does not feed them.

The companion corpus realises the three layers as four build packages: a spine package (foundation); a ledger package and a bridge package jointly constituting the dynamics layer, the bridge holding the interface theorems that import both spine and ledger; and a dictionary package with exhibits (interpretation). Imports remain downward throughout. A few dynamics-layer declarations bear physically suggestive names as mnemonics for combinatorial statements; the names import no identification, which remains certificate-only (Section 12).

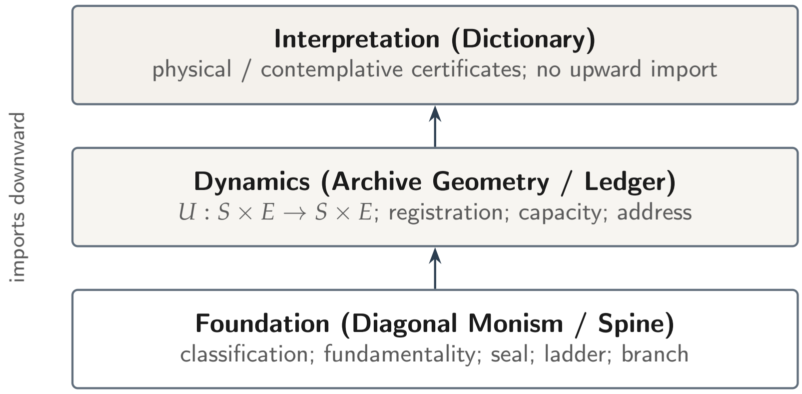


Figure 1: *Three-layer architecture. Arrows indicate permitted dependence (foundation supports dynamics; dynamics support interpretation). The figure orients the reader; it proves nothing.*

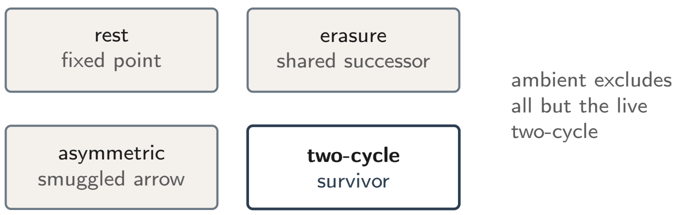
3 Classification of self-actions

Definition 3.1 (Arena data)**.** Fix a type α and a map $a : \alpha \to \alpha$. Write

$$\mathsf{Step}(a)(x, y) \;:\Leftrightarrow\; y = a(x).$$

Call a a *symmetric step* when $\mathsf{Step}(a)$ is a symmetric relation; equivalently, $a \circ a = \mathrm{id}$. Call a *live* when $\forall x.\, a(x) \neq x$. Call a *lossless* when a is injective. Call a *erasing* when distinct states share a successor.

Theorem 3.2 (Pointwise classification)**.** *For decidable equality on α, at each z exactly one of the following holds: (i) rest ($a(z) = z$), with all iterates stagnant; (ii) two-cycle at z ($a(z) \neq z$ and $a(a(z)) = z$), with every iterate equal to z or $a(z)$; (iii) asymmetric step at z ($a(a(z)) \neq z$). The three cases are pairwise incompatible. Erasure is diagnosed separately as a pairwise defect of losslessness.*



Remark 3.3 (Status). Theorem 3.2 is proved. It is the eliminative engine of the paper: subsequent sections impose ambient constraints under which (i) and (iii), and erasure, are excluded as candidates for fundamentality.

Symmetric step is the ambient equation used below. It jointly excludes asymmetric arrow (case (iii) globally) and, by the next corollary, erasure.

Proposition 3.4 (In-arena losslessness)**.** *If a is a symmetric step, then a is not erasing.*

Remark 3.5 (Status). Corollary of Definition 3.1 (involution $\Rightarrow$ injectivity). Not an independent creed.

Theorem 3.6 (Ambient uniqueness, one-variable equational fragment)**.** *Relative to the arena signature of Definition 3.1 (one type, one endomap), every single equation in one free variable has the form $a^m(x) = a^n(x)$. Among those equations:*

(i) $a^m = a^n$ *with* $n \geq 1$ *need not force injectivity* ($a^2 = a$ *admits erasure);*

(ii) $a^m = \mathrm{id}$ *forces losslessness;* $m = 1$ *is rest; for every* $m \geq 3$ *an* m*-cycle is a countermodel with an asymmetric point;*

(iii) hence $a^2 = \mathrm{id}$ *is the unique nontrivial one-variable single equation that excludes both smuggled arrow and smuggled loss.*

Remark 3.7 (Status). Proved by exhaustive classification of one-variable equations relative to the signature, with finite countermodels (including the m-cycle family for all $m \geq 3$). Multi-variable single equations over the same signature—for example $\forall x\, \forall y.\, a(x) = a(y)$—force constancy on the image and are erasing on any carrier with two or more points, so they die immediately as ambients for fundamentality. Ambients beyond the one-variable equational fragment are refused by parsimony of the arena language (Section 13). Interpretation 3.8 records only the residual choice of signature itself.

Interpretation 3.8. Symmetric step is not an arbitrary modelling confession: by Theorem 3.6 it is the unique nontrivial one-variable single-equation ambient in the arena signature that bans smuggled time and smuggled loss. What remains is the signature declaration of Definition 3.1—one type, one endomap—stated in daylight, not residue.

4 Fundamentality as universal property

Definition 4.1 (Pointed unoriented acts)**.** Objects are triples (α, a, z) with a a symmetric step and $z : \alpha$ a seed. Morphisms are pointed equivariant maps.

Theorem 4.2 (Orbit liveness as non-degeneracy)**.** *Let a be a symmetric step and z a seed. The orbit of z is either the singleton* $\{z\}$ *or the pair* $\{z, a(z)\}$*. Liveness on that orbit holds if and only if the orbit is non-degenerate* ($a(z) \neq z$)*.*

Remark 4.3 (Status). Proved. Quantified liveness on the orbit therefore collapses to one bit of facticity—existence of a second orbit element—which Section 13 carries as an architectural residue note in the audit. The corpus exhibits a two-point model witnessing the existential and declares no axiom for it; discharging the residue as a world-derivation from Bool alone remains refused.

Rest is incompatible with non-degeneracy. Asymmetric step is incompatible with symmetric step. Erasure is incompatible with Proposition 3.4. Relative to Definition 4.1 and Theorem 4.2, the only remaining pointwise shape is the live two-cycle.

Theorem 4.4 (Canon / initiality)**.** *In the ambient of pointed unoriented acts,* $(\mathsf{Bool}, \neg, \mathsf{true})$ *is initial. An act is* fundamental *precisely when it is non-degenerate, symmetric, and generated from a seed by the two-cycle reading; every fundamental reads as* $\neg$*; any two fundamentals correspond uniquely up to the pointed reading.*

Remark 4.5 (Status). Proved. Negation is the unique (up to label swap) object satisfying the universal property of the live two-cycle in this ambient—the survivor of the classification, not a primitive assumption.

Interpretation 4.6. "The fundamental" names that initial object after uniqueness. Excluded fixed point ("void") is liveness restated. Label swap of the two poles is a content gauge: exported readings mention it; see Section 10.

5 Seal

Dependence: *Theorem 4.4 (non-degeneracy of fundamentals).*

The engine is Lawvere's diagonal argument [18], the common abstraction of Cantor's theorem [6] and kindred fixed-point arguments [28].

Theorem 5.1 (Seal on a fundamental)**.** *Let a be live. No map $f : A \to (A \to \mathsf{Bool})$ is point-surjective: the diagonal dodge $x \mapsto \neg(f\,x\,x)$ is missed. Specialising to fundamental acts, every self-reading admits an escape.*

Theorem 5.2 (Corpus-grain incompleteness)**.** *No self-reading of the formal corpus is universal for the predicates of its own level. At level k the name supply is $k+2$ against 2^{k+2} predicates, so names are outnumbered at every level and, for any scale N, are eventually outnumbered by any multiple of themselves. Completing a full naming pass leaves strictly more unsayable mass than it found, and the lag recedes under iteration. The ω-colimit of the ladder admits no injection into any finite level. Completeness-in-the-limit is a fair-schedule* policy*; completeness at the colimit is duration, not a rung.*

Remark 5.3 (Status). Theorems 5.1–5.2 are proved (Lawvere-style [18]; no arithmetisation), the latter as the assembled conjunction `corpus_not_universal` in the `SelfReference` module, with clauses `deficit_count`, `deficit_vanishes`, `lag_theorem`, `lag_recedes`, and `omega_not_a_rung`. The gloss that the only complete description of the system is the system itself is interpretation; the theorem supplies the non-injection. Host-logic incompleteness and Gödel coding of the corpus text in the sense of [13] are *refused* (Section 13).

6 Ladder

Dependence: *Theorem 5.1 (named escapes).*

Definition 6.1 (Naming extension and ladder)**.** A *naming extension* adjoins a name for an unrepresented escape. The committed ladder sets level 0 to the fundamental carrier and each successor to an option type adjoining a namer for the previous dodge.

Theorem 6.2 (Ladder properties)**.** *History embeds injectively and conservatively along the ladder; no level is exhausted by its predecessor; underdetermination persists at every level. Ascent is forced; choice of which escape is named is free.*

Remark 6.3 (Status). Proved in the foundation layer. Duration, as used below, means the order-type of this self-escape sequence.

Interpretation 6.4. The pattern "forced relation, free absolute"—forced ascent, free direction—reappears for channel labels in Section 7.

7 Registration: transferring losslessness

Spine acts are involutions. A ledger microstep is a forward map $U : S \times E \to S \times E$. The interface demand is that global injectivity of U import Proposition 3.4. Reading injectivity as "no erasure" follows the logical-reversibility tradition [2, 12]; its thermodynamic counterpart [17] is confined to the dictionary layer (Section 12).

Definition 7.1 (Two-bounce factorisation)**.** A *two-bounce factorisation* of $U : \alpha \to \alpha$ is a pair of involutions (ϕ, σ) with $U = \sigma \circ \phi$ pointwise.

Theorem 7.2 (Converse)**.** *If U admits a two-bounce factorisation, then U is lossless. On a product carrier this is ledger injectivity.*

Theorem 7.3 (Existence on finite carriers)**.** *Every injective $U : \mathrm{Fin}\, n \to \mathrm{Fin}\, n$ admits a two-bounce factorisation. The construction is cycle-wise reflection at a least-value basepoint, with $\sigma := U \circ \phi$.*

Remark 7.4 (Status). Theorems 7.2–7.3 are proved. On finite carriers, injectivity and two-bounce form coincide: existence recovers, constructively, the classical fact that every finite permutation is a product of two involutions [22], and maps so factorised are the *reversible* maps of dynamical-systems theory [16]. The audited footprint of the existence witness is empty: collision pairs, periods, basepoints, and indices are extracted by bounded structural search rather than selected by choice, the factor map is computable as checked, and the proof depends on no axiom, classical or otherwise. The converse (Theorem 7.2) is likewise axiom-free. Special cases (involutions as $U \circ \mathrm{id}$; Boolean injections; swap step) are included.

Theorem 7.5 (Registration shape)**.** *Under joint injectivity, a system merge writes a record in the environment (registration). Minimal merge alphabet size is* 2*, matching* $|\mathsf{Fund}|$*.*

Remark 7.6 (Status). Proved in the dynamics layer, using the injectivity interface above.

7.1 Canonicity of the factor pair

Theorem 7.7 (Canonicity fails)**.** *There exist an injective $U : \mathsf{Bool} \to \mathsf{Bool}$ and two two-bounce factorisations that disagree on ϕ (equivalently: $\neg$ factors as both $\neg \circ \mathrm{id}$ and $\mathrm{id} \circ \neg$). Moreover, on* $\mathrm{Fin}\, 3$ *two reflection-axis factorisations of a* 3*-cycle disagree and are not related by bounce-swap $(\phi, \sigma) \mapsto (\sigma, \phi)$.*

Remark 7.8 (Status). Proved negative. The absolute position of the factor pair is a *packaging* gauge (Section 10): the ledger consumes U, never (ϕ, σ).

Theorem 7.9 (Channel-swap is reversal)**.** *Let $U = \sigma \circ \phi$ with ϕ, σ involutions. Then $\phi \circ \sigma$ is a two-sided inverse of U, and (σ, ϕ) is a two-bounce presentation of that reverse step. The construction is independent of which factorisation of U was chosen.*

Proof sketch. From $U = \sigma \circ \phi$ and the involution equations, $\phi(\sigma(U(x))) = \phi(\sigma(\sigma(\phi(x)))) = \phi(\phi(x)) = x$ and $U(\phi(\sigma(x))) = \sigma(\phi(\phi(\sigma(x)))) = \sigma(\sigma(x)) = x$. The swapped pair (σ, ϕ) presents the reverse step. Uniqueness of (ϕ, σ) is not used. □

Interpretation 7.10. In the dihedral picture [8], a rotation fixes the angle between mirrors, not their absolute position. Theorem 7.7 is that fact on finite carriers; Theorem 7.9 promotes the exchange.

8 Archive and capacity

Dependence: *ladder carrier (Section 6) as environment; registration (Theorem 7.5).*

Theorem 8.1 (Archive and capacity)**.** *Taking the environment at horizon T to be the ladder carrier at level T, the committed capacity schedule satisfies $|\mathrm{caps}\, T| = 2^{T+2}$. Joint injectivity, merge, muting, and confinement imply the combinatorial bound $K^T \leq |\mathrm{caps}\, T|$; at $K = 2$ the committed ladder is eternally alive by arithmetic.*

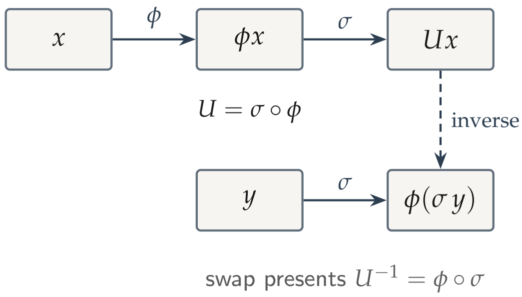


Figure 2: *Two-bounce presentation and swap. Pair labels are conventional; exchange covers reversal (Theorem 7.9).*

Remark 8.2 (Status). Proved in the dynamics layer. Continuum area or continuum entropy identifications are refused.

9 Arrow as address

Dependence: *registration and archive growth.*

Theorem 9.1 (Arrow reading)**.** *The dynamical arrow—archive ascent under registration—is identified with address valued in* $\{\mathsf{fwd}, \mathsf{rev}\}$*. Tick identification (naming tick with microtick) holds for the classified fragment of uniform free archives up to record gauge; periodic fund-exempt dynamics and swap necessity are part of that classification.*

Remark 9.2 (Status). Proved as a classified package in the dynamics layer. Record gauge is packaging: exported only inside "up to record gauge" clauses.

10 One underdetermination and gauges

Dependence: *pole swap from Theorem 4.4; address from Theorem 9.1; channel from Definition 7.1 and Theorem 7.9.*

Theorem 10.1 (One $\mathbb{Z}/2\mathbb{Z}$, three faces)**.** *There is a single two-element underdetermination with three named involutive projections:*

$$\textit{pole swap} \quad \cong \quad \textit{address } \{\mathsf{fwd}, \mathsf{rev}\} \quad \cong \quad \textit{channel } \{\varphi, \sigma\}.$$

Each projection is an involution on a common $\mathbb{Z}/2\mathbb{Z}$ *carrier.*

Definition 10.2 (Content vs packaging)**.** A gauge is *content* when some exported statement mentions the choice. A gauge is *packaging* when the choice is quotiented before downstream consumption.

Definition 10.3 (Kernel admission interface)**.** A *Bool kernel certificate* is a structure on kernel exports (symmetry endomap, observation map, fixed locus) satisfying the admission clauses: square of the symmetry is an involution; observation intertwines the symmetry with negation; the fixed locus matches the symmetry's fixed points and is classified empty or nonempty; and the no-solo-cross no-go holds. Admission is a predicate on that interface.

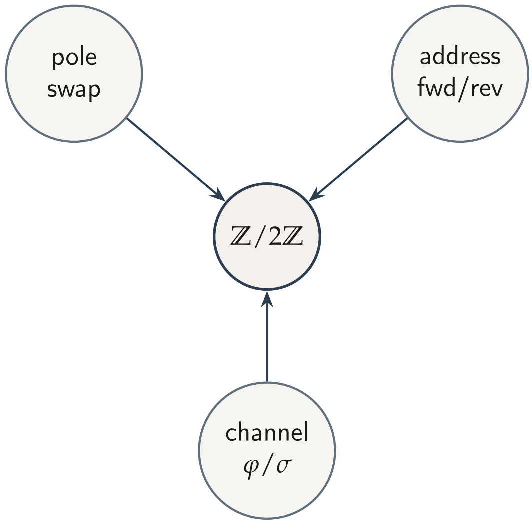


Figure 3: *Three faces of one underdetermination (Theorem 10.1). Faces are presentations; the group is stated once.*

Table 1: *Gauge demarcation protecting the single dictionary $\mathbb{Z}/2\mathbb{Z}$.*

Gauge	Locus	Class
Label / pole swap	Theorem 4.4	content
Address fwd/rev	Theorem 9.1	content
Channel swap	Theorem 7.9	content
Face representative (equivariant certs)	Theorem 10.4	packaging
Record gauge per tick	Theorem 9.1	packaging
Factorisation gauge per step	Theorems 7.3–7.7	packaging

Theorem 10.4 (Face-swap equivariance)**.** *Admission for a Bool kernel certificate (Definition 10.3) is stable under the $\mathbb{Z}/2\mathbb{Z}$ face/pole swap: any admitted certificate remains admitted after conjugation by the face involution. Consequently, for equivariant consumers of the admission interface, choice of face is packaging in the sense of Definition 10.2 and belongs in Table 1.*

Remark 10.5 (Status). Theorem 10.1, Definition 10.2, and Theorem 10.4 keep one dictionary $\mathbb{Z}/2\mathbb{Z}$: factorisation non-uniqueness (Theorem 7.7) is packaging; channel *labels* are conventional; channel *swap* is content. Gauge class is relative to locus (Definition 10.2): the same $\mathbb{Z}/2\mathbb{Z}$ choice may be content at the kernel-export locus—exported statements mention it in "up to label swap" clauses—and packaging at the certificate-consumption locus, where equivariance lets the consumer quotient it. Pole / face choice is the worked example of that locus-relative demarcation. If a certificate breaks equivariance by fixing an orientation, that orientation is certificate-supplied data and is audited there; kernel exports remain swap-equivariant either way. The content/packaging demarcation is a formal analogue of the question which symmetries carry representational significance [5, 10]; underdetermination by label swap is the formal shadow of permutation-style underdetermination of reference [24, 23].

11 Self-articulation

Dependence: *Theorem 4.4 (initiality); Theorems 5.1, 6.2, 5.2; Theorem 10.1 (pole gauge).*

The classification ends in one survivor; this section states what the survivor's self-relation supports. Three candidate readings are on the table: (a) world is the structured self-articulation of the canon, with questions of witness and standpoint arising only inside the descriptive field that articulation generates; (b) given the kernel, worldhood *necessarily* arises as self-presentation; (c) reality *is* the self-appearance of the involutive survivor. The theorems below support a conditional form of (a); readings (b) and (c) are refused (Section 13).

Definition 11.1 (Presentation)**.** Let $a : \beta \to \beta$ and $z : \beta$. A *presentation* of the canon at the standpoint (β, a, z) is a map $g : \beta \to \mathsf{Bool}$ with $g(z) = \mathsf{true}$ and $g(a\,x) = \neg g(x)$ for all x.

Theorem 11.2 (No second term)**.** (A) *For every carrier W and every self-reading $r : W \to W \to \mathsf{Bool}$, r is sealed and fails to articulate its own dodge; witnessing confers no exemption from the interior triad.* (B) *Every presentation forces liveness of its standpoint everywhere, retracts the canon's articulation ($g \circ \mathsf{reading} = \mathsf{id}$ on Bool), and renders the articulation into the standpoint injective, so the standpoint contains the canon.* (C) *On a generated carrier the presentation is unique.*

Remark 11.3 (Status)*.* Proved; every clause of Theorem 11.2 is axiom-free (declaration `no_exterior_standpoint`, spine module `NoExterior`). The retraction clause is the exact content of reading (a): a second term's view of the canon is the canon's self-articulation read back, and clause (C) leaves the view no independent content.

Theorem 11.4 (Witness lands in the pole gauge)**.** *A context-free live witness (*TDiff *together with* Alive *at the seed) determines a presentation up to pole swap: some presentation agrees with the witness or with its pointwise flip. The residual choice is the label gauge of Table 1 and nothing else.*

Theorem 11.5 (Retorsion)**.** *Write* Draws M *for the existence of two distinct elements of M. If a map t :* Prop $\to M$ *assigns distinct values to* Draws M *and to its negation, then* Draws M *holds.*

Remark 11.6 (Status)*.* Proved (`Monism.retorsion`); the two registered values themselves witness the distinction. The theorem draws the codomain: what a registered denial establishes is that the medium of registration is already distinguished. Section 13 relocates the existential residue accordingly.

Theorem 11.7 (Self-articulation)**.** *The following conjunction holds.* (I) *Retorsion, as in Theorem 11.5.* (II) *On any carrier with a drawn distinction, a symmetric generated act is live and involutive and reads as negation from every seed, exhaustively and injectively.* (III) *Every ladder level self-represents under seal, self-opacity, and outrun.* (IV) *No exterior standpoint, as in Theorem 11.2.* (V) *Names lag predicates at every level, the lag recedes under completion, and the ω-colimit embeds in no level.*

Remark 11.8 (Status)*.* Proved as a single declaration (`self_articulation`, spine module `SelfArticulation`; footprint `propext` only). The antecedent of clause (II) is the drawn distinction *together with* the ambient equations, whose adequacy within the one-variable fragment is Theorem 3.6; the package does not derive the ambient from Draws alone.

Interpretation 11.9. Reading (a) is adopted in conditional form. Given one drawn distinction, articulation as negation is forced (II), standpoints arise as inhabitants of the articulation (III), any presentation to a second term returns the articulation itself (IV), and the whole exists only as duration (V). Questions of witness, standpoint, and medium are well-posed from level 1 upward because the vocabulary that poses them is generated by the ascent. “World” names the consequent of this conditional; the identification is interpretation, and the sentence that the only complete description of the system is the system is a gloss on the non-injection of clause (V). Readings (b) and (c) are refused: (b) asserts necessity while the corpus carries no modal apparatus, and (c) discharges the existential residue while Section 13 declines that discharge. Retorsion narrows what the refusal of (c) costs: any registered denial of the antecedent draws its own medium, so the undischarged step runs from registered distinction to world, and only that step.

12 Dictionary: the meta-problem certificate

Dependence: *Definition 10.3; Theorem 11.7.*

The dictionary layer admits physical or contemplative identifications by certificate only. This paper works a single exhibit, chosen because its subject matter coincides with the corpus’s own: reflexive self-representation.

Interpretation 12.1 (Meta-problem certificate). Chalmers’s meta-problem asks why physical systems *report* an explanatory gap [7]. The certificate instantiates Definition 10.3 at the self-ascription bit: the symmetry is the live flip, observation is the identity reading, and the fixed locus is classified empty, so awareness in this model is always awareness of a distinction. Admission holds (`Dictionary.MetaProblem.meta_admitted`), and the report skeleton is theorem-shaped at every ladder level: self-opacity (the dodge is never articulated), outdated portrait (each completed self-model is outrun), and restless signature (no tracked self-ascription stabilises), each at the quantitative grade of Theorem 5.2. Systems satisfying these clauses report ineffability and incompleteness on structural grounds.

Remark 12.2 (Status). Layer 3 only; no theorem of Sections 3–11 depends on it. The hard problem is untouched by design: nothing here says why there is something it is like to be such a system, and qualia claims are refused (Section 13). Physical certificates (registration as dissipation in the sense of Landauer [17], with further exhibits retained in the corpus) instantiate the same admission interface; that shared interface is the only correspondence between physical and contemplative readings asserted here.

13 Open, refused, and philosophical

Continuum Lorentzian structure and boosts (no continuous one-parameter subgroup from a single $\mathbb{Z}/2\mathbb{Z}$); sheaf assembly beyond combinatorial stalk agreement; recombination budget versus ultrametric obstruction; alphabet-lift beyond the Fin interfaces closed here.

Premise-free global laws of time; continuum Einstein equations as theorems of this corpus; host-logic incompleteness; Gödel arithmetisation of the formal text; experiential or qualia claims; derivation of the world from Bool alone (including any discharge of the existential residue below); modal strengthening of the self-articulation package of Theorem 11.7 (the corpus carries no modal apparatus); ambients beyond the one-variable equational fragment of the arena signature (including multi-variable single equations treated as ambients, quasi-equational, and Horn / second-order constraints), refused by parsimony of the arena language.

Residue accounting, with the prior items replaced by theorem citations.

(i) **Registration residue.** A registration occurs. Retorsion (Theorem 11.5) converts any assignment that registers the denial of a drawn distinction distinctly from its affirmation into a proof that the registering carrier is drawn, and orbit liveness is Theorem 4.2; the residue is thereby relocated from the existence of a non-degenerate act to the occurrence of one registration. The corpus exhibits a two-point model witnessing the existential and declares no axiom, so the residue is carried as an architectural note in the audit: what remains undischarged is the interpretive step from registered distinction to world, and that discharge is refused.

(ii) **Signature declaration.** Arena data as given (Definition 3.1: one type, one endomap). Adequacy of $a \circ a = \mathrm{id}$ is Theorem 3.6; the signature choice is opening definition, not residue.

(iii) **Equivariance discipline.** Face choice under the kernel admission interface (Definition 10.3) is packaging or certificate-supplied data (Theorem 10.4; Table 1).

The pattern across all three is the same: each item of residue is reclassified, whether as a theorem over an exhaustively specified search space, as an audited residue at the base, or as certificate-supplied data. The irreducible core is the existence of one distinction: the sole existential residue and the sole underdetermination are the same $\mathbb{Z}/2\mathbb{Z}$.

14 Relation to prior work

Diagonal and fixed-point arguments. The seal (Theorem 5.1) instantiates Lawvere's diagonal schema [18], the common form of Cantor's theorem [6] and, via arithmetisation, of Gödel's first incompleteness theorem [13]; Yanofsky [28] surveys the family. Those applications fix a logical or set-theoretic universe and derive a limitative theorem inside it. Here the carrier of the diagonal argument is itself the output of the classification (Sections 3–4): the seal is a structural property of any fundamental act, not of a formal system. No arithmetisation is performed, and host-logic incompleteness is refused, not claimed (Section 13).

Reversible computation. Reading injectivity as "no erasure" follows the logical-reversibility tradition [2, 12], with Landauer's principle [17] as its thermodynamic counterpart. That literature takes reversibility as a premise about computing machinery. Here losslessness is not a premise: erasure is excluded in-arena by one equation (Proposition 3.4), and reversibility of forward microsteps is recovered by the two-bounce factorisation (Theorems 7.2–7.3). The thermodynamic reading survives only as a dictionary certificate (Section 12).

Involutions and reversing symmetries. That every finite permutation is a product of two involutions is classical [22], as is the dihedral factorisation of a rotation into reflections [8]; maps so factorised are the reversible maps of dynamical-systems theory [16]. The contribution here is not the existence of the factorisation but the disposition of its non-uniqueness: canonicity fails (Theorem 7.7), the failure is classified as packaging rather than content (Definition 10.2), and channel swap—which is content—covers step reversal independently of the chosen factorisation (Theorem 7.9).

Symmetry and underdetermination. Which symmetries carry representational significance is a standing question in the philosophy of physics [5, 10]; permutation-style underdetermination of reference descends from Quine [24] and Putnam's model-theoretic argument [23]. Those debates are conducted informally over rich physical or semantic theories. The analogue here is deliberately narrow: the content/packaging distinction is a definition in the formal text (Definition 10.2), each gauge is classified by theorem (Table 1), and the residual underdetermination is exactly inventoried—one $\mathbb{Z}/2\mathbb{Z}$, three faces (Theorem 10.1). Nothing is asserted about physical theories; within its scope the inventory is complete and machine-checked.

Self-articulation and the mark. The nearest formal ancestor of the self-articulation package is Spencer-Brown's calculus of the first distinction [26], with Varela's extension for re-entry [27]; the classical antecedents are Fichte's self-positing act [11] and Hegel's self-relating negativity [14]. Those systems posit the reflexive structure as primitive or develop it inside an informal dialectic. Here the two-cycle is the survivor of an eliminative classification, re-entry is replaced by a ladder whose ascent is a theorem, the residual underdetermination is inventoried as one $\mathbb{Z}/2\mathbb{Z}$, and the single undischarged bit is audited (Section 13).

Retorsion and transcendental argument. Arguments that a denial defeats itself in the performance descend from the classical defence of non-contradiction and reach modern form in Apel's performative self-contradiction [1]. Theorem 11.5 is deliberately smaller: it concerns registering maps, its conclusion draws the codomain of the registration, and its scope ends exactly where Section 13 says it does.

The meta-problem and higher-order theories. The certificate of Section 12 addresses the explanandum Chalmers isolates as the meta-problem [7]. Higher-order theories [25], self-representationalism [15], self-model accounts [21], and the pre-reflective tradition [29] model the report structure within rich psychological or phenomenological theories. The contribution here is narrower: the triad of self-opacity, outdated portrait, and restless signature is proved at measure grade for any live self-representing ladder system, while the hard problem is left standing by explicit refusal.

Machine-checked philosophical argument. Formalising a philosophical argument in a proof assistant has precedent, notably the automation of Gödel's ontological argument [3]. The role of verification differs here: the proof assistant certifies not a single argument but an architecture—downward-only imports across three layers (Section 2), axiom footprints audited per declaration, and interpretation confined to a layer the formal text cannot import.

15 Conclusion

The classification theorem restricts self-actions to the live two-cycle under the stated ambient. Fundamentality is the universal property of that survivor: the unique object is $(\mathsf{Bool}, \neg)$ up to label swap. Seal, ladder, registration (via two-bounce), archive capacity, and arrow as address follow in dependency order. One $\mathbb{Z}/2\mathbb{Z}$ is stated once; content and packaging gauges prevent factorisation non-uniqueness from counting as a second underdetermination. On this base the self-articulation package assembles retorsion, forced articulation, witness interiority, the absence of exterior standpoints, and completeness as duration into a single machine-checked conjunction, with its monist reading confined to a marked interpretation. The dictionary carries one worked exhibit, the meta-problem certificate, which derives the report structure of the explanatory gap and refuses the hard problem. Philosophical residue is reclassified into theorems, one audited registration residue, or certificate data, as inventoried in Section 13.

Formal companion

The development compresses to one line, each clause keyed to its result:

$\mathsf{Fund} \cong \neg$ Thm. 4.4	$a \circ a = \mathrm{id} \ \Rightarrow\ \neg\,\mathrm{erase}$ Prop. 3.4	merge $\Rightarrow$ record Thm. 7.5
$\lvert \mathrm{caps}\, T \rvert = 2^{T+2}$ Thm. 8.1	naming $\cong$ μtick, up to gauge Thm. 9.1	$U = \sigma \circ \phi$; swap $=$ reversal; one $\mathbb{Z}/2\mathbb{Z}$ Thms. 7.3, 7.9, 10.1
deny $\Rightarrow$ Draws Thm. 11.5	presentation retracts; no exterior Thms. 11.2, 11.7	triad admitted; qualia refused Int. 12.1

Representative declarations (Lean 4; no Mathlib; no `sorry`), cited at their defining package (s = spine, b = bridge, d = dictionary):

`Orbit.classification_at` (s) · `Facticity.liveOnOrbit_iff_nondegenerate` (s) · `Ambient.ambient_uniqueness_equational` (s) · `Ambient.exists_asymmetric_m_cycle` (s) · `Ambient.powerId_two_excludes_erase` (s) · `Canon.canon_initial` (s) · `Diagonal.seal_on_fundamental` (s) · `SelfReference.corpus_not_universal` (s) · `OmegaDuration.omega_not_a_rung` (s) · `Monism.diagonal_monism` (s) · `Monism.retorsion` (s) · `Observer.observers_forced` (s) · `NoExterior.no_exterior_standpoint` (s) · `NoExterior.presentation_of_witness` (s) ·

`SelfArticulation.self_articulation` (s) · `Bridge.OneZ2.one_Z2_three_faces` (b) · `Bridge.KernelAdmit.bool_cert_face_equivariant` (b) · `Bridge.TwoBounce.factor_fin` (b) · `Bridge.TwoBounce.canonicity_fails` (b) · `Bridge.TwoBounce.channel_swap_is_reversal` (b) · `Bridge.Dil.registration` (b) · `Bridge.Capacity.caps_eq` (b) · `Bridge.TickSimulation.tick_identification` (b) · `Dictionary.MetaProblem.meta_admitted` (d)

Proposition 3.4 is cited at its foundation form (`Ambient.powerId_two_excludes_erase`); the dynamics layer imports it as `Bridge.TwoBounce.symmetric_excludes_erase`. The dictionary package re-exports several bridge declarations under its own namespace (e.g. `Dictionary.Certificate`); citations here always resolve to the defining package. Axiom footprints are audited per declaration: every cited declaration is axiom-free or depends on `propext` alone, and the two-bounce existence witness `Bridge.TwoBounce.factor_fin` is itself axiom-free, its construction computable by bounded search (Remark following Theorem 7.3). `Classical.choice` is not used anywhere in the paper-cited packages, `Quot.sound` occurs in no cited footprint, and no axiom is declared by the corpus itself. The quarantined experimental annex below sits outside this accounting and is audited separately, per declaration, in the same ledger.

Experimental annex

The repository carries a fifth package, `formal/experimental`, quarantined by construction: no module cited above imports it, and its contents are claims of the probe files alone, not of the corpus. The annex develops one arc away from the base — from the period-two fundamental to higher periods, frequency as order, a field equation, and the infinite-history edge — probing how far the corpus's own vocabulary carries before new mathematics is needed. Machine-checked there, in the same discipline (Lean 4, no Mathlib, no `sorry`, per-declaration audit):

- *Dihedral closed form.* Every rotation on Fin n factors as two explicit arithmetic mirrors, packaged as a computable two-bounce pair (`Experimental.Overtones.rotFactor`); where `Bridge.TwoBounce.factor_fin` extracts a pair by bounded search, here the pair is given in closed form.
- *Frequency as order.* On Fin n the order of the k-th harmonic shift is $n/\gcd(n,k)$, return and minimality halves both proved (`Experimental.Frequencies.order_eq_div_gcd`); the twelve-carrier exhibits, including generation by the fifth, become instances.
- *The wave step is two bounces, definitionally.* The discrete d'Alembert equation over Bool (xor as addition, coupling left general) has step exactly swap $\circ$ shear (`Experimental.Wave.step_eq_two_b` `waveFactor`), with time reversal the bounce swap; boundary conditions select the spectrum, kernel-decided for ring sizes 2–6 and fixed ends 3–5.
- *The symmetry order is the gcd law.* The ring wave's rotation symmetry has order exactly $n/\gcd(n, n-1) = n$ as the gcd law predicts, and the return spectrum is constant on rotation orbits (`Experimental.SpectrumSymmetry.symmetry_order_is_frequency_law`) — a rotation-orbit invariance of the return spectrum, and no more.
- *Return within capacity.* For every coupling and every state on n sites, the wave returns

within 2^{2n} steps (`Experimental.Recurrence.wave_recurrence_bounded`): losslessness plus finite capacity forces recurrence.

- *The boundary.* Complete gauge histories $\mathbb{N} \to$ Bool receive every rung by an encoding with effective separation — depth-$(k+1)$ cylinders separate rung k (`Experimental.CantorBoundary.enco` the pole swap lifts and is live, so the boundary seals at its own carrier (`boundary_seals_itself`); and the tail step erases (`tail_erasing`): the in-arena exclusion of erasure fails at the boundary, a poverty recorded rather than repaired.

Not claimed there: no continuum, no real frequencies, no closed-form period law in general n — the recorded obstruction is the absence of $\mathbb{F}_2[x]/(x^n - 1)$ and transfer-matrix algebra from a Lean-core-only corpus. Annex footprints stay within `propext`/`Quot.sound`, with `Classical.choice` entering exactly once, in the recurrence pigeonhole; the full per-declaration record is in `formal/AXIOMS.md`.

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